Design and Dimensioning of a PEM Fuel Cell Stack for a Delivery Van

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***Abstract*—Proton exchange membrane fuel cells (PEMFCs) are promising power sources for zero-emission transportation due to their high power density, rapid startup, and excellent dynamic response. This study presents the design, dimensioning, and performance analysis of a 32 kW PEM fuel cell stack intended for urban delivery van applications. Material selection is optimized for automotive cost–performance balance.**

An electrochemical framework is outlined to estimate hydrogen consumption using Faraday's law and to evaluate the open-circuit voltage through the Nernst equation. Major electrochemical losses, including activation, ohmic, and concentration overpotentials, are discussed using the Butler–Volmer formulation and transport considerations. In addition, a system-level mass and energy balance is developed to examine the thermal behaviour of the stack and to motivate the choice of an appropriate cooling strategy.

The results demonstrate that the proposed PEMFC stack provides a compact and technically feasible power solution for urban electric mobility, particularly when integrated with battery hybridization to handle transient load demands. As an exploratory step toward physics-based modelling, the AlphaPEM simulation [10][11] framework was briefly examined to gain preliminary insight into the modelling approaches used to describe PEM fuel cell behaviour.

***Index Terms*—PEM, Fuel Cells**

I. Introduction

THE increasing demand for sustainable transportation has driven the development of hydrogen fuel cell technologies for automotive applications. Proton Exchange Membrane fuel cells (PEMFCs) are well suited for urban vehicles due to their low operating temperature, high power density, and fast response to changing load conditions. In this project, a PEM fuel cell stack is designed and dimensioned to meet the power requirements of an urban delivery van. The study includes material selection, stack sizing, and evaluation of electrochemical performance using fundamental relations such as Faraday's law, the Nernst equation, and the Butler–Volmer equation. In addition, fuel consumption, efficiency, and thermal management are analyzed through mass and energy balance calculations.

II. Stack design and dimensioning

PEMFCs operate at 60–80 ℃ with fast startup (<5 min), high power density (>1 kW/L), and excellent dynamic response – ideal for stop-and-go urban driving. High-temperature fuel cells (SOFC 800–1000 ℃, PAFC 160–220 ℃) unsuitable due to slow warmup and heavy insulation.

Advantages: high power density, compact stack, battery hybridization, quick refueling. Limitations: ultra-pure H_2 required (CO poisons Pt at 10 ppm), water management challenges, PFSA/PTFE regulatory risk.

A. Cell Materials

Materials selected for automotive cost-performance balance:

- Membrane: Nafion PFSA (20–50 µm), 0.1 S/cm at 80 ℃ – industry standard, proven durability
- Catalyst: Pt/C (cathode 0.3–0.4 mg/cm^2, anode 0.05 mg/cm^2) – 3× cathode loading for slower ORR, meets DOE 2025 cost targets
- GDL+MPL: Toray carbon paper + PTFE – mass-producible, low cost
- Bipolar plates: Stainless steel + carbon coating (0.15 mm/plate) – 5× thinner than graphite, 30% lighter, stamped production reduces cost to <20 €/kW vs 50 €/kW for graphite

B. Fuel and Storage

Pure compressed H_2. PEMFCs do not accept CO/reformate impurities, so no onboard reforming. DMFC too low power density and methanol crossover issues. Tanks: 350/700 bar composite cylinders (current FCEV standard). Size for $\sim$300 km urban range based on stack efficiency + van consumption. Fits chassis limits per delivery van demos.

C. Stack Sizing

Operating point: $U_{\text{cell}} = 0.65$ V, $j = 1.0$ A/cm^2, $A_{\text{active}} = 250$ cm^2.

Cell power:

$$P_{\text{cell}} = 0.65 \times 1.0 \times 250 = 162.5 \text{ W}. \tag{1}$$

Stack power target 32 kW (meets average delivery van traction power):

$$N_{\text{cells}} = \frac{32000}{162.5} = 196.92 \approx 200 \text{ cells}. \tag{2}$$

Nominal voltage:

$$V_{\text{stack}} = 200 \times 0.65 = 130 \text{ V}. \tag{3}$$

This design matches commercial automotive stacks (e.g., Ballard FCvelocity 30 kW modules) and provides sufficient power for urban delivery cycles when combined with battery for acceleration peaks.

D. Nernst OCV (80 °C, 2 bar H_2/air)

$$E_0(353) = 1.229 - 0.00085 \times (353 - 298) = 1.182 \text{ V},$$

$$\frac{RT}{2F} = \frac{8.314 \times 353}{2 \times 96485} = 0.0152 \text{ V},$$

$$\ln\left(\frac{2 \times \sqrt{0.42}}{0.47}\right) = 1.014,$$

$$E_{\text{OCV}} = 1.182 + 0.0152 \times 1.014 = 1.197 \approx 1.20 \text{ V/cell}.$$

$$StackOCV : 200 \times 1.20 = 240 \; V(theoretical).$$

E. Physical Dimensions

Cell pitch 1.8 mm (MEA 0.1 mm, 2×GDL 0.4 mm, 2×BP 0.3 mm, seals 0.3 mm, coolant 0.4 mm). Active length: $L_{\text{active}} = 200 \times 1.8 = 360$ mm. Total length (end plates, manifolds, compression): 450–500 mm.

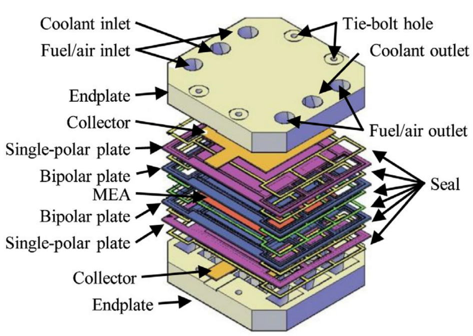

Fig. 1. PEMFC stack assembly showing bipolar plates, MEA, seals, end plates, manifolds and tie bolts.

F. Materials and Dimensions Impact on Performance

Current density $j = 1.0$ A/cm^2 enables compact stack (volumetric power density ∼1.2 kW/L) but reduces efficiency to $\eta \sim 50\%$ vs 60% at 0.5 A/cm^2 due to higher activation and ohmic losses.

Thin membrane (30 µm) reduces ohmic resistance (∼0.05 Ωcm^2) but increases H_2 crossover (∼2 mA/cm^2), reducing OCV and efficiency.

Metallic bipolar plates improve power density (3× vs graphite) and reduce cost but require durable coatings to prevent corrosion-induced performance degradation over 5000 h lifetime.

G. Trade-offs and Limitations

32 kW stack meets delivery van average power needs (battery handles peaks). Design feasibility validated by commercial automotive stacks with similar specifications.

Limitations: H_2 infrastructure availability, catalyst degradation under load cycling (∼10% voltage loss over 5000 h), PFAS regulatory restrictions.

Mitigations: battery hybridisation smooths load cycles (reduces degradation), advanced water management (prevents flooding/drying), alternative membrane materials under development (hydrocarbon ionomers).

III. Performance Analysis

A. Fuel Requirement at Various Load Conditions

According to Faraday's law, the molar flow rate of hydrogen is, $I = zFr$

- *Stack Current*

$$I_{\text{Stack}} = \frac{P_{\text{Stack}}}{U_{\text{Cell}} \times N_{\text{Cells}}} \tag{4}$$

$$I_{\text{Stack}} = \frac{32000}{0.65 \times 200} = 246.15 \, A$$

- *Hydrogen Flow Rate*

$$r_{H_2} = \frac{I_{\text{Stack}}}{zF} \tag{5}$$

$$r_{H_2} = \frac{246.15}{2 \times 96485} = 0.001275 \, mol/s$$

- *Hydrogen Mass Flow Rate*

Hydrogen molar mass:

$$\dot{m}_{H_2} = r_{H_2} \times M_{H_2}, M_{H_2} = 2.016 \, g/mol$$

$$\dot{m}_{H_2} = 0.001275 \times 2.016 = 0.00257 \, g/s$$

- *Hydrogen Required for Different Load Conditions*

Load (%)	Stack Power (kW)	Stack Current (A)	H_2 mass flow (g/s)
50	16	123	0.00129
75	24	185	0.00193
100	32	246	0.00257

B. Open Circuit Voltage Using Nernst Equation

The Nernst equation for a PEM fuel cell is:

$$E_{Cell} = E^\circ + \frac{RT}{2F} \ln\left(\frac{P_{H_2}\sqrt{P_{O_2}}}{P_{H_2O}}\right) \tag{6}$$

Where:

$$E^\circ = 1.229 \, V$$

$$E^\circ = -\frac{\Delta G^\circ}{nF} = \frac{-237130}{2 \times 96485} = 1.229\,V$$

Assume:

$$T = 353\,K, P_{H_2} = 1\,atm, P_{O_2} = 0.21\,atm, P_{H_2O} = 1\,atm$$

$$E_{Cell} = 1.229 + \frac{8.314 \times 353}{2 \times 96485} \ln\left(\frac{1 \times \sqrt{0.21}}{1}\right)$$

$$E_{Cell} = 1.229 + 0.0152(-0.78) = 1.217\,V$$

- *Stack Open Circuit Voltage*

$$E_{Stack} = E_{Cell} \times N_{Cells} \tag{7}$$

$$E_{Stack} = 1.217 \times 200 = 243.4\,V$$

C. Butler–Volmer Equation

$$j = j_0 \left[\exp\left(\frac{\alpha z F \eta}{RT}\right) - \exp\left(-\frac{(1-\alpha) z F \eta}{RT}\right)\right] \tag{8}$$

First exponential $\rightarrow$ oxidation reaction
Second exponential $\rightarrow$ reduction reaction
Assume typical PEMFC values:

$$\alpha = 0.5, \quad z = 1, \quad T = 353K, \quad \eta = 0.10V$$

$$\frac{\alpha z F \eta}{RT} = \frac{0.5 \times 96485 \times 0.1}{8.314 \times 353} = 1.64$$

$$\exp(1.64) = 5.15$$

$$\exp(-1.64) = 0.19$$

The smaller exponential contributes less than 4%, therefore it can be neglected:

$$j = j_0 \exp\left(\frac{\alpha z F \eta}{RT}\right)$$

Taking logarithm:

$$\ln j = \ln j_0 + \frac{\alpha z F \eta}{RT}$$

$$\eta_{act} = \frac{RT}{\alpha z F} \ln\left(\frac{j}{j_0}\right)$$

Given:

$$j = 1.0\,A/cm^2 and j_0 = 10^{-3}\,A/cm^2$$

$$\eta_{act} = \frac{8.314 \times 353}{0.5 \times 96485} \ln(1000) = 0.42V$$

Load Case	Current Density (A/cm^2)	η_{act} (V)
a	0.5	0.38
b	0.75	0.41
c	1.0	0.42

D. Activation Loss at Different Loads

- *Ohmic Loss*

Ohmic losses arise from:

- Proton resistance of the membrane
- Electronic resistance of electrodes and bipolar plates

$$\eta_{ohm} = j R_{area}$$

Membrane thickness:

$$L = 50\mu m$$

Proton conductivity:

$$\sigma = 10\,S/m$$

$$R_{mem} = \frac{L}{\sigma}$$

$$R_{mem} = \frac{5 \times 10^{-5}}{10} = 5 \times 10^{-6} \Omega m^2$$

Including contacts and plates:

$$R_{mem} = 2.0 \times 10^{-5} \Omega m^2$$

Current Density (A/cm^2)	η_{ohm} (V)
0.3	0.06
0.6	0.12
1.0	0.20

- *Concentration Loss*

Occurs near limiting current density.

$$\eta_{conc} = -\frac{RT}{zF} \ln\left(1 - \frac{j}{j_L}\right) \tag{9}$$

Assume:

$$j_L = 2.0 A/cm^2$$

Current Density (A/cm^2)	η_{conc} (V)
0.3	0.005
0.6	0.010
1.0	0.021

E. Polarization and Power–Current Density Curves

The polarization curve of the PEM fuel cell stack is calculated using Eq. (10), which describes the stack voltage as a function of current density by combining the reversible Nernst voltage with the activation, ohmic, and concentration overpotentials [1].

$$V_{\text{stack}}(j) = N\left[E_{\text{cell}} - \eta_{\text{act}}(j) - \eta_{\text{ohmic}}(j) - \eta_{\text{conc}}(j)\right] \quad (10)$$

where $V_{\text{stack}}(j)$ denotes the stack voltage as a function of current density. N is the number of cells connected in series in the stack, and E_{cell} is the reversible voltage of a single cell obtained from the Nernst equation under the given operating conditions. The terms η_{act}, η_{ohmic}, and η_{conc} represent the activation, ohmic, and concentration overpotentials, respectively.

Figure 3 combines the polarization and power–current characteristics of the PEM fuel cell stack in a single plot using dual y-axes. The polarization curve shows that the stack voltage decreases as the current density increases due to electrochemical (activation and concentration) and resistive (ohmic) losses. In parallel, the power–current (power-density) curve illustrates how the stack power density varies with current density. It initially increases as current rises, reaches a maximum value, and then decreases because the voltage drop caused by polarization losses becomes dominant at higher current densities.

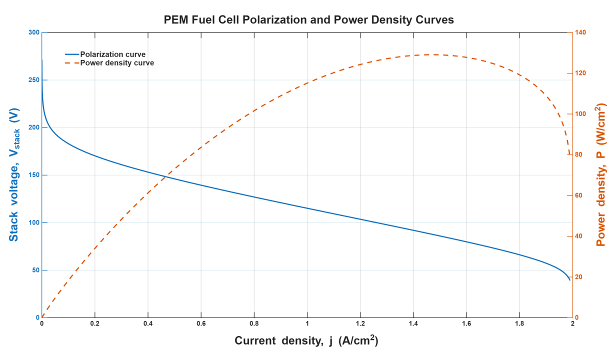

Fig. 2. Polarization and Power - Current Density Curves

F. Fuel Cell Efficiencies

- *Thermodynamic Efficiency*

$$\varepsilon_{th} = \frac{\text{Maximum electrical work}}{\text{chemical energy}} = \frac{\Delta G}{\Delta H} \quad (11)$$

For hydrogen oxidation, the maximum electrical work is $\Delta G = 237$ kJ/mol. Using HHV ($\Delta H = 286$ kJ/mol) and LHV ($\Delta H = 242$ kJ/mol):

$$\varepsilon_{th} = \frac{237}{286} = 0.83 = 83\% \quad @HHV$$

$$\varepsilon_{th} = \frac{237}{242} = 0.98 = 98\% \quad @LHV$$

- *Voltage Efficiency*

$$\varepsilon_U = \frac{U}{U^0} = \frac{0.65}{1.0} = 65\%$$

- *Faradaic Efficiency*

$$\varepsilon_F = \frac{I/zF}{\dot{n}_{in,fuel}} \quad (12)$$

$$\varepsilon_F = \frac{250/(2 \times 96485)}{1.35 \times 10^{-3}} = 0.96$$

- *Overall Efficiency*

$$\varepsilon_{cell} = \varepsilon_{th} \times \varepsilon_U \times \varepsilon_F$$

$$\varepsilon_{cell} = 0.83 \times 0.65 \times 0.96$$

$$\varepsilon_{cell} = 0.517 = 52\%$$

To explore physics-based modelling of PEM fuel cells, the stack was simulated using the AlphaPEM framework[10][11]. The simulation produced polarization and power–current density curves for the 32 kW stack. While these results are not fully optimized, they demonstrate how a physics-based approach can capture the behaviour of the stack under different loads.

With further refinement and calibration using experimental data, this modelling approach could be extended and optimized to represent realistic, real-world operating conditions for a delivery van PEMFC system.

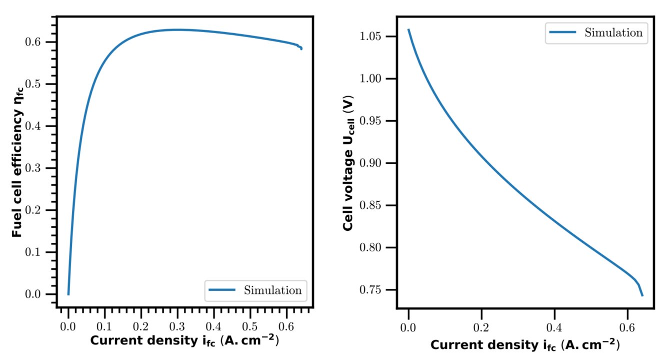

Fig. 3. voltage and cell efficiency versus current density from AlphaPEM simulation.

IV. Mass and Energy Balance, Cooling

A. Mass Balance

Using Faraday's law, the hydrogen molar consumption rate of the fuel cell stack can be calculated from the stack current and the number of cells.

$$\dot{n}_{H_2,cons} = N \times (I/(2F)) \quad (13)$$

Where,

$$N = 200 \quad and \quad I = 246.15A$$

$$\dot{n}_{H_2,cons} = 200 \times (246.15/192970)\dot{n}_{H_2,cons} = 0.255117 mol/s$$

The molar flow rate is converted to mass flow rate using the molar mass of hydrogen.

$$\dot{m}_{H_2} = 0.2551174 \times 2.016 \times 10^{-3} Kg/mol$$

$$\dot{m}_{H_2} = 0.0005143166 Kg/s$$

Convert to kg/h:

$$\dot{m}_{H_2} = 0.0005143166 \times 3600 = 1.85154 Kg/h$$

B. Energy Balance

The PEM fuel cell stack converts the chemical energy of hydrogen into electrical energy, with the remaining energy appearing as heat. At steady state, the energy balance is:.

$$E_{\text{chem}} = P_{\text{stack}} + Q_{\text{gen}} \tag{14}$$

Here, $\dot{E}_{\text{chem}}$ is the chemical power input, $\dot{P}_{\text{stack}}$ is the electrical power output, and $\dot{Q}_{\text{gen}}$ is the heat generated in the stack. Using the molar hydrogen flow rate from Faraday's law, the chemical input is:

$$E_{\text{chem}} = \dot{m}_{H2} \cdot LVH_{\text{H2}} \tag{15}$$

For our 32 kW stack:

$$\dot{m}_{H2} = 0.5143 \text{ g/s}, \quad LVH_{\text{H2}} = 120 \text{ kJ/g}$$

$$E_{\text{chem}} \approx 61 \text{ KW}$$

Heat Loss from the System: The portion of chemical energy that is not converted into useful electrical power appears as waste heat:

$$Q_{\text{gen}} = E_{\text{chem}} - P_{\text{stack}} = 61 \text{ kW} - 32 \text{ kW} = 29 \text{ kW}$$

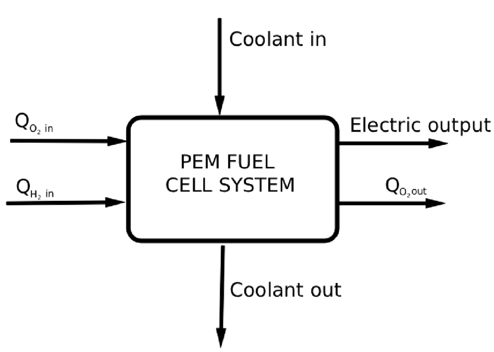

Fig. 4. PEMFC stack energy balance

C. Required Cooling

To maintain the stack at 70–80 ℃, this heat must be removed. Using a liquid coolant (water or water-glycol) with $c_p = 4.18 \text{ kJ/(kg K)}$ and allowing a temperature rise of $\Delta T = 10$ K:

$$\dot{m}_{\text{coolant}} = \frac{Q_{\text{gen}}}{c_p \Delta T} \approx 0.69378 \text{ kg/s}$$

PEM fuel cells (70–80 ℃) reject significant waste heat (40–60% of input energy) [5][9], and inadequate cooling leads to membrane dehydration, thermal stresses, and performance degradation. Reported cooling strategies include heat spreaders, air cooling, liquid cooling, and nanofluids [5]-[7][9]. Heat spreaders are suitable only for small stacks ($<$10 kW). Air cooling is simple but limited by low heat capacity and high parasitic losses, making it viable mainly for low-power systems ($<$5 kW)[5][9]. Liquid cooling using water or 50/50 water–ethylene glycol in a radiator–pump loop with active control is the dominant solution for automotive and mid/high-power stacks[7][8][9]. Nanofluids offer modest heat exchanger size reduction but negligible overall performance improvement relative to added complexity.

The 32 kW stack rejects 29 kW of heat, requiring continuous removal to maintain 70–80 ℃ operation. Based on the calculated requirement ($\dot{m} \approx 0.72 \text{ kg s}^{-1}$ for $\Delta T = 10 \text{ K}$), air cooling lacks sufficient capacity, heat spreaders are inadequate, and nanofluids provide limited benefit.

Therefore, consistent with established practice, a closed-loop liquid cooling system using a 50/50 water–ethylene glycol mixture is selected as the most technically appropriate and industrially validated solution.

The system that we propose consists of a pump–radiator–reservoir loop with a bypass valve and PI-based temperature control with load feed-forward. A 50/50 water–ethylene glycol mixture is chosen for high heat capacity, freeze protection, corrosion resistance, and proven reliability, providing an optimal balance of performance, cost, and maturity.

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